

Shayne Magstadt

Curriculum Vitae

Ph.D., Colorado State University, Department of Forest and Rangeland Stewardship
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Education

Ph.D. 2026, Forest Sciences, Colorado State University – Fort Collins, CO
M.S. 2021, Natural Resources: Environmental Science and Management, Cal Poly Humboldt – Arcata, CA
B.S. 2015, Fish, Wildlife, and Conservation Biology (Minor in Spatial Information Management), Colorado State University – Fort Collins, CO

Teaching Experience

Spring 2021

Instructor, Colorado State University, Warner College of Natural Resources, Fort Collins, CO – GIS Applications in Natural Resource Management

Spring 2022 – Spring 2023

Instructor, Colorado State University, Warner College of Natural Resources, Fort Collins, CO – Applications of Global Positioning Systems

Research and Professional Experience

June 2026 – Present

Postdoctoral Fellow, Conduct research at the intersection of artificial intelligence, geospatial science, remote sensing, and wildland fire management. Develop computer vision, machine learning, and cellular automata models to simulate wildfire spread, assess water-drafting suitability, and analyze spatial and operational data relevant to wildfire response. Integrate imagery, terrain, fuels, weather, and other geospatial datasets to improve fire-behavior modeling and support more effective decision-making during wildfire planning and suppression.

January 2023 – Present

Volunteer Firefighter, Rist Canyon Volunteer Fire Department, Bellvue, CO – RCVFD is an all-volunteer, rural fire department which provides fire suppression services (wildland – primarily initial attack, structural, vehicle and other fire), Basic Life Support (BLS) Emergency Medical Services, Light rescue and extrication and other ‘general’ emergency response in our response area as defined in the RCVFD by-laws. RCVFD’s primary goal is safety of RCVFD responders in all cases – hence any response or tactic will be mitigated by that primary goal. In this role I have helped to successfully apply for federal, state, and private grants to acquire firefighting equipment and training funds. I am a member of the grants committee, our mapping task force.

Spring 2020 – Spring 2026

Graduate Research Assistant, Colorado State University, Fort Collins, CO – Spearheaded innovative machine learning projects focused on enhancing forest restoration and wildfire management strategies. Developed predictive models using artificial neural networks to identify areas requiring hazardous fuel treatments across Colorado's national forests. Conducted comprehensive analysis on firefighter retention, employing an extensive dataset from interagency hotshot crews to uncover key factors influencing workforce stability. Developed a cutting-edge mapping technique for the real-time tracking and continuous monitoring of aerial water drops during wildfire suppression efforts.

Fall 2018 – Spring 2020

Graduate Research Assistant, Cal Poly Humboldt, Arcata, CA – Led comprehensive field data collection efforts focused on understanding the ecological impacts of black bear-induced redwood bark stripping events. Executed precise, tree-level measurements and gathered high-resolution aerial imagery for advanced analytical studies, contributing to a deeper understanding of forest health dynamics. Innovated by developing a sophisticated model to quantify individual tree damage from black bears, integrating high-resolution LiDAR point clouds with hyperspectral UAV imagery. This model offers insights into wildlife behavior and its effects on forest ecosystems.

Spring 2015 – Spring 2018

Foothills Wildlife Research Facility, Colorado Parks and Wildlife, Fort Collins, CO – Played a role in a cutting-edge research project aimed at implementing adaptive management practices to mitigate the spread of plague among small mammal populations. Executed field operations including the trapping, anesthesia administration, and venipuncture of small mammals to obtain blood samples for detailed laboratory analysis. Administered vaccines through precision-guided, GPS-equipped pellet distributors mounted on ATVs, enhancing the efficiency and reach of vaccination efforts. Organized the compilation, management, and meticulous organization of extensive GPS data, derived from both handheld units and the advanced vaccine distribution system, to inform strategic decision-making. Contributed to the scientific community by generating maps and figures, which were featured in publications such as the *Journal of Wildlife Diseases*, thereby advancing knowledge in wildlife disease management.

Peer Reviewed Publications

1. **Magstadt, S.**, Wadhwani, R., Whitley, M. A., Martin, T. J., Sitter, E., Wei, Y., & O'Connor, C. D. 2026. A high spatial resolution daily fire perimeter progression dataset for wildfires in the Western United States: 2020–2024. *Scientific Data*.
2. **Magstadt, S.**, Belval, E.J., Pietruszka, B.M., Wei, Y. and O'Connor, C., 2026. An assessment of aerial firefighting response times between agencies during the 2020 fire season in California. *International Journal of Disaster Risk Reduction*, p.106105.
3. **Magstadt, S.**, Wei, Y., Pietruszka, B.M., Calkin, D.E., 2024. A Deep Learning Approach for Predicting Aerial Suppressant Drops in Wildland Firefighting Using Automatic Dependent Surveillance–Broadcast Data. *Fire*, 7(11), 380.
4. **Magstadt, S.**, Wei, Y, 2025. A Neural Network Model to Study Factors Impacting the Selection of Hazard Fuel Treatment Types in Colorado's National Forests. *International Journal of Wildland Fire. Journal of Wildland Fire*, 34(1).
5. Belval, E.J., Bayham, J. and **Magstadt, S.**, 2024. Retention of highly qualified wildland firefighters in the Western United States. *Forest Policy and Economics*, 158, p.103115.
6. **Magstadt, S.**, Gwenzi, D. and Madurapperuma, B., 2021. Can a Remote Sensing Approach with Hyperspectral Data Provide Early Detection and Mapping of Spatial Patterns of Black Bear Bark Stripping in Coast Redwoods?. *Forests*, 12(3), p.378.
7. Madurapperuma, B., Lamping, J., McDermott, M., Murphy, B., McFarland, J., Deyoung, K., Smith, C., MacAdam, S., Monroe, S., Corro, L. and **Magstadt, S.**, 2020. Factors Influencing Movement of the

Manila Dunes and Its Impact on Establishing Non-Native Species. Remote Sensing, 12(10), p.1536.

Contributed Presentations

1. Functional trait variation among plant functional types in the Canadian Arctic tundra. ABOVE 10th Science Team Meeting, 21st May 2024, Boulder, Colorado
2. Trends of growing season productivity in the Arctic tundra near Inuvik, Northwest Territories, CA. ABOVE 10th Science Team Meeting, 21st May 2024, Boulder, Colorado
3. A Neural Network Model to Study Factors Impacting the Selection of Hazard Fuel Treatment Types in Colorado's National Forests. The 19th Symposium on Systems Analysis in Forest Resources (SSAFR 2022), Estes Park, Colorado
4. Surveying coastal dune ecosystems using multi-spectral drone imagery in the North Spit of Humboldt Bay. INRSEP/CNRS Research Symposium, 4th October 2019, Library, Humboldt State University.
5. UAV photogrammetry for surveying dune habitats: A review of research needs of the Friends of the Dunes Land Trust. INRSEP/CNRS Research Symposium, 28th September 2018, Library, Humboldt State University.
6. Trends and Applications in Wildland Fire Behaviour Modelling and Simulation from 2000–2025. 10th International Conference on Forest Fire Research, Coimbra, Portugal.

Research Grants

1. Advancing technology for environmental monitoring in geospatial science courses. Colorado State University, Warner College of Natural Resources, \$3,495.00 (Funded 2024). Duran S., Tulanowski E., Magstadt S.

Non-Research Grants

1. Woodward, \$5,000.00, general funding, funded December 2022
2. DFPC (Direct Disbursement), \$74,900.00, 10 SCBAs, funded October 2023
3. Rotary Club of Fort Collins Breakfast, \$3,500.00, medical supplies, funded December 2023
4. El Pomar Foundation, \$10,000.00, PPE, funded January 2024
5. Gary Sinise Foundation, \$50,000.00, Zoll X-Series and 2 SCBAs, funded March 2024
6. Larimer County Fire Grant Program, \$10,000.00, two land pumps, funded March 2024
7. Larimer County Community Mitigation Grant Program, \$2,500.00, tablets and lasers, funded April 2024
8. CSFFF, \$500.00, general funding, funded May 2024
9. CDPHE, \$24,189.66, Zoll X-Series and 4 AEDs, funded September 2024
10. DFPC FFSDP, \$11,553.00, three thermal imaging cameras, funded March 2025
11. Larimer County Fire Grant Program, \$3,952.55, Mule litter, funded April 2025
12. Proper Twelve Grant, \$10,000.00, cordless extrication tool, funded July 2025
13. CL Fires, \$1,000.00, laptop, funded August 2025

14. FEMA AFG, \$45,714.28, 10 units bunker gear, funded September 2025

15. Larimer County Fire Grant Program, \$7,917.00, floating portable pump, funded April 2026

Non-Peer Reviewed Publications and Other Contributions

1. Tripp, D. W., Corro, L. M., **Magstadt, S. R.**, & Sack, D. A. (2018). Plague Management Techniques and Monitoring in Colorado's Prairie and Shrub-steppe Ecosystems. Colorado Parks and Wildlife.

Accolades

Wildfire Partner Program Ambassador	2024
Rist Canyon Volunteer Fire Department Mapping Task Force	2024
Rist Canyon Volunteer Fire Fighter	2023
Rist Canyon Volunteer Fire Department Grants Committee	2023
Xi Sigma Pi Honor Society, Colorado State University	2015

Certifications

Red Card Certification	2024
Part 107 Remote Pilot Certificate	2019
PADI Open Water Diver Certificate	2007

Relevant Skills and Interests

Software and Programming: Python, Artificial Intelligence and Machine Learning (deep learning, computer vision, geospatial modeling), TensorFlow, R, JavaScript, Google Earth Engine, Microsoft Planetary Compute, Anaconda, ArcGIS Pro, QGIS, ENVI, SQL, GIMP, Google Earth Pro, DJI Ground Station Pro, DJI GO/GO4, Mission Planner, Adobe Photoshop and Illustrator, PuTTY

Hardware and Equipment: DJI Phantom 4, DJI Phantom 3, DJI Matrice 100, Handheld GPS, Trimble GPS, Emlid Reach RS+, Total Station, MicaSense RedEdge Sensor, ATV and 4WD Operation, Radio Telemetry

Personal Interests and Hobbies: Hiking, Hunting, Fishing, Glass Blowing, Woodworking, Camping, Sustainability, Backpacking, Gardening, automated/AI-based irrigation and plant monitoring (IoT)